My Homepage

Zeyu Zheng

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EXPERIENCES

ByteDance Seed May 2025 - Sep 2026

Research Scientist Intern, Top Seed Talent Program

- **Mid-training**: Developed data synthesis pipelines (Socratic dialogues, conjecture–resolution paper pairing) for Seed 2.5, consistently improving Math-SimpleQA accuracy and reducing Proof-PPL.
- **Reasoning post-training**: Conducted RL algorithm ablations for Seed 2.1; delivered 200k+ RL prompts for large-scale Seed Evolving training, improving ResearchMath scores from ∼16% to ∼54%.
- **Evaluation**: Led the development of ResearchMath, used in 30k+ internal evaluation trails across Seed Evolving and Seed 2.5; developed Math-SimpleQA and Proof-PPL benchmarks for Seed 2.5 mid-training evaluation.
- **Coding post-training**: Developed end-to-end Lean-SWE post-training pipelines (infrastructure, data and reward design) using GitHub-synthesized data for Seed 2.0, improved performance on internal autoformalization and formal theorem-proving benchmarks.
- **Formal math research**: Co-led BFS-Prover-V2 (infrastructure, data and training ablations); designed agent harnesses and worked on agentic post-training for Seed-Prover 1.5.

NVIDIA NeMo Sep 2026 - Oct 2026

Research Scientist Intern

- Developed RL and evaluation infrastructure for NeMo Gym (merged PR, 2000+ LoC), and evaluated open-source models for data synthesis.
- Delivered 50k+ RL prompts for Nemotron training, improving performance on MathArena and boosting arXivMath accuracy from ∼43% to ∼55%.

EDUCATION

Carnegie Mellon University Pittsburgh, PA Sep 2023 -

Ph.D. candidate in the ACO program, affiliated with ICARM and L3 Lab

Advisor: Prasad Tetali & Sean Welleck

Fudan University Shanghai, China Sep 2019 - Jun 2023

B.S. in Mathematics, Top-notch Talent Program in Mathematics

SELECTED PUBLICATIONS

* indicates equal contributors and names are listed in alphabetical or random order

7. Zeyu Zheng, Yiyang Liu, Yuyang Zhao, Congmin Zheng, Jiachen Zhu, Jianian Chen, Kangning Zhang, Kayue Daniel Wong, Lingyu Yang, Rong Shan, Shihao Chen, Thomas Zhu, Wentao Zhang, Xiaopeng Li, Yan Hu, Yongxi Lin, Yuxiang Chen, Zhuoying Ou, Jianghao Lin, Jeremy Avigad, Prasad Tetali, Jingbang Chen, and Sean Welleck, *ReuF2F: Benchmarking coding agents on formalized undergraduate research problems*, under review. GitHub preprint
6. Michail Sarantis*, Prasad Tetali*, and Zeyu Zheng*, *On Counting Independent Sets in Uniform Hypergraphs*, under review. arXiv
5. Zeyu Zheng*, Shengtong Zhang*, Jeremy Avigad, Prasad Tetali, and Sean Welleck, *The Problem Is the Problem: Towards Scalable Mathematical Discovery*, MATH-AI @ NeurIPS 2026, also under reivew at conference. arXiv
4. ByteDance Seed, Seed Foundation Models Seed2.0(Model Card) Seed2.1(Model Card)
3. ByteDance Seed, Seed Prover series GitHub Seed-Prover 1.5

2. Ran Xin*, Zeyu Zheng*, Yanchen Nie*, Kun Yuan, and Xia Xiao, *BFS-Prover V2: Scaling up Multi-Turn Off-Policy RL and Multi-Agent Tree Search for LLM Step-Provers*, ICML 2026.arXiv

1. Boris Bukh*, Ting-wei Chao*, and Zeyu Zheng*, *The Oddtown problem modulo a composite number*, Combinatorics, Probability and Computing. arXiv

SERVICES

Conference reviewer: AAAI, ARR, EMNLP, ICLR, ICML, KDD, NeurIPS, SODA, STOC, WWW
Journal referee: Combinatorics Probability and Computing, Journal of Graph Theory, Mathematics of Computation, SIAM Journal on Discrete Mathematics

FULL PUBLICATIONS

* indicates equal contributors and names are listed in alphabetical or random order

21. Zeyu Zheng, Yiyang Liu, Yuyang Zhao, Congmin Zheng, Jiachen Zhu, Jianian Chen, Kangning Zhang, Kayue Daniel Wong, Lingyu Yang, Rong Shan, Shihao Chen, Thomas Zhu, Wentao Zhang, Xiaopeng Li, Yan Hu, Yongxi Lin, Yuxiang Chen, Zhuoying Ou, Jianghao Lin, Jeremy Avigad, Prasad Tetali, Jingbang Chen, and Sean Welleck, *ReuF2F: Benchmarking coding agents on formalized undergraduate research problems*, under review. GitHub preprint

20. Michail Sarantis*, Prasad Tetali*, and Zeyu Zheng*, *On counting Independent Sets in Uniform Hypergraphs*, under review. arXiv

19. Zeyu Zheng*, Shengtong Zhang*, Jeremy Avigad, Prasad Tetali, and Sean Welleck, *The Problem Is the Problem: Towards Scalable Mathematical Discovery*, MATH-AI @ NeurIPS 2026, also under reivew at conference. arXiv

18. ByteDance Seed, *Seed2.1 Model Card: Agentic Intelligence for Productivity*, technical report. Homepage Model Card

17. Jiachen Zhu, Zhuoying Ou, Congmin Zheng, Yuxiang Chen, Zeyu Zheng, Rong Shan, Lingyu Yang, Lionel Z Wang, Weiwen Liu, Yong Yu, Weinan Zhang, and Jianghao Lin, *Contexting as Recommendation: Evolutionary Collaborative Filtering for Context Engineering*, under review. arXiv

16. Rong Shan, Te Gao, Hang Zheng, Yunjia Xi, Jiachen Zhu, Zeyu Zheng, Yong Yu, Weinan Zhang, and Jianghao Lin, *Academic Conferences are Potentially Facing Denominator Gaming Caused by Fully Automated Scientific Agents*, ICML 2026. arXiv

15. Weihua Du, Jingming Zhuo, Yixin Dong, Andre Wang He, Weiwei Sun, Zeyu Zheng, Manupa Karunaratne, Ivan Fox, Tim Dettmers, Tianqi Chen, Yiming Yang, Sean Welleck, *AdaExplore: Failure-Driven Adaptation and Diversity-Preserving Search for Efficient Kernel Generation*, Transactions on Machine Learning Research (TMLR). arXiv

14. Chenyu Zhou, Huacan Chai, Wenteng Chen, Zihan Guo, Rong Shan, Yuanyi Song, Tianyi Xu, Yingxuan Yang, Aofan Yu, Weiming Zhang, Congming Zheng, Jiachen Zhu, Zeyu Zheng, Zhuosheng Zhang, Xingyu Lou, Changwang Zhang, Zhihui Fu, Jun Wang, Weiwen Liu, Jianghao Lin, and Weinan Zhang, *Externalization in LLM Agents: A Unified Review of Memory, Skills, Protocols and Harness Engineering*, Transactions on Machine Learning Research (TMLR). arXiv

13. Jiachen Zhu, Lingyu Yang, Rong Shan, Congmin Zheng, Zeyu Zheng, Weiwen Liu, Yong Yu, Weinan Zhang, Jianghao Lin, *Turing Test on Screen: A Benchmark for Mobile GUI Agent Humanization*, KDD 2026. arXiv

12. ByteDance Seed, *Seed2.0 Model Card: Towards Intelligence Frontier for Real-World Complexity*, technical report. Homepage Model Card

11. ByteDance Seed AI4Math, *Seed-Prover 1.5: Mastering Undergraduate-Level Theorem Proving via Learning from Experience*, technical report. GitHub arXiv

10. Ran Xin*, Zeyu Zheng*, Yanchen Nie*, Kun Yuan, and Xia Xiao, *BFS-Prover V2: Scaling up Multi-Turn Off-Policy RL and Multi-Agent Tree Search for LLM Step-Provers*, ICML 2026. arXiv

9. Boris Bukh*, Ting-wei Chao*, and Zeyu Zheng*, *The Oddtown problem modulo a composite number*, under review. arXiv

8. Anurag Bishnoi*, Bartłomiej Kielak*, Benedek Kovács*, Zoltán Lóránt Nagy*, Gábor Somlai*, Máté Vizer* and Zeyu Zheng*, *The generalized trifference problem*, IEEE Transactions on Information Theory (2026). arXiv Journal version

7. Congmin Zheng*, Jiachen Zhu*, Zhuoying Ou*, Yuxiang Chen, Kangning Zhang, Rong Shan, Zeyu Zheng, Mengyue Yang, Jianghao Lin, Yong Yu, and Weinan Zhang, *A Survey of Process Reward Models: From Outcome Signals to Process Supervisions for Large Language Models*, ACL 2026. arXiv

6. János Barát*, Zoltán L. Blázsik*, Balázs Keszegh*, and Zeyu Zheng*, *On the number of edges in saturated partial embeddings of maximal planar graphs*, Discrete Mathematics. arXiv

5. Tibor Jordán*, Robin Huang*, Henry Simmons*, Kaylee Weatherspoon*, and Zeyu Zheng*, *Four-regular graphs with extremal rigidity properties*, Discrete Mathematics 347 (2024), no. 4, 113833. EGRES Journal version

4. Logan Post* and Zeyu Zheng*, *Common kings of a chain of cycles in a strong tournament*, Graphs and Combinatorics 39 (2023), no. 4, 71. arXiv Journal version

3. Ruinian Chang*, Jingbang Chen*, Ian Munro*, Richard Peng*, Qingyu Shi*, and Zeyu Zheng*, *Sandpile Prediction on Undirected Graphs*, under review. arXiv

2. Ervin Győri*, Xianzhi Wang*, and Zeyu Zheng*, *Extremal planar graphs with no cycles of particular lengths*, Discrete Mathematics. arXiv

1. Chaoliang Tang*, Hehui Wu*, Shengtong Zhang*, and Zeyu Zheng*, *On the Turán number of the linear 3-graph C_{13}*, The Electronic Journal of Combinatorics (2022), P3–46. arXiv Journal version